

AKSHAT PAL

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PROFESSIONAL SUMMARY

Software Engineer specializing in **web development**, **cloud computing**, **AI/ML**, and **process automation**. Experienced in end-to-end **design** and delivery of scalable software solutions, including automation workflows built with UiPath and **cloud deployments on AWS**. Strong foundation in **artificial intelligence**, **robotics**, and **cryptography**, with a track record of building applied **IoT**, **computer-vision**, and automation systems from concept to deployment.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java

Web Development: HTML5, CSS3, JavaScript, Node.js, Flask

Cloud & DevOps: AWS Cloud Services, Git, GitHub

AI / Machine Learning: Artificial Intelligence, Machine Learning, Computer Vision (YOLOv5)

Automation & RPA: UiPath, Process Automation, Web Automation

Embedded & Robotics: Embedded Systems Design, Autonomous Drones & Robotics

Core CS: Data Structures & Algorithms, Database Management, Cryptography

Soft Skills: Problem Solving, Critical Thinking, Adaptability, Communication, Interpersonal Skills, Reasoning, Organizational Skills, Detail-Oriented

EDUCATION

Vellore Institute of Technology, Chennai, India

2023 – Present

Bachelor of Technology, Computer Science Engineering — Specialization in Artificial Intelligence and Robotics

Delhi Public School, Bhopal, India

Graduated 2022

Awarded 8 Scholar Badges, Star Badge, Scholar Trophy, Maroon Blazer (for scoring above 85% academically for 7 consecutive years), and 2 Full Attendance Trophies

EXPERIENCE

Freelance Software Engineer — Fiverr, Upwork, Outlier.ai, and other platforms

July 2026 – Present

- Develop web applications, automation workflows, and custom software solutions for clients across global freelance platforms.
- Deliver scalable, efficient, and user-focused solutions tailored to diverse business requirements.

Summer Intern — Maulana Azad National Institute of Technology, Bhopal

June 2025

- Developed the Vitals Driven Home Automation System with Sleep Vigilance System, a smart sleep-monitoring and home automation platform using ESP32 and ESP8266 microcontrollers.
- Built and trained a custom AI model for sleep-stage prediction and anomaly detection, designed for paralyzed and hospitalized patients.

PROJECTS

Vitals Driven Home Automation System with Sleep Vigilance System *IoT, AI & ML, ThingSpeak, Twilio, Flask*

- Built a smart sleep-monitoring and home automation system on ESP32/ESP8266 with a custom-trained AI model for sleep-stage prediction and anomaly detection, tailored for paralyzed and hospitalized patients.

Local Tourism Website

HTML5, CSS3, JavaScript, Node.js

- Designed a responsive tourism website showcasing destinations across Tamil Nadu, India, built to be easily adaptable to any other location or region.

YouTube Transcript Summarizer Launcher Extension

JavaScript

- Built a Chrome extension providing instant access to YouTube video transcripts and AI-powered summarization using the user's preferred AI platform.

Drone-Based Fire Detection System

YOLOv5, AWS Cloud, Computer Vision, Python

- Developed an AI-enabled early-alert system for forest-fire detection combining drone-based imagery with scalable, cloud-based (AWS) processing.

Automated Product Price Comparison

UiPath, Web Automation, AI APIs, Data Processing

- Built an AI-driven price-comparison platform delivering real-time, personalized product recommendations, eliminating manual cross-platform shopping.
- Used UiPath RPA for live e-commerce web scraping, Streamlit for interactive data visualization, and the Google Gemini API to generate consumer insights.

PATENTED PROJECT

System and Method for Proximity-Based Audio Generation

Application No. 202641054231

- Developed a contactless audio-generation framework transforming spatial interactions into real-time musical output through intelligent sensing and adaptive signal processing.
- Enhanced human-machine interaction by enabling intuitive, touch-free engagement with responsive, immersive audio experiences.

ADDITIONAL COURSES & CERTIFICATIONS

- Master C++ Programming — GeeksforGeeks
- OCI GenAI Professional — Oracle